

Carson Rivera

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EDUCATION

Florida State University

Tallahassee, FL

Bachelor of Science in Computer Science - Honors Program | GPA: 3.88/4.00

May 2027

EXPERIENCE

Software Engineer Intern

Jun. 2026 – Aug. 2026

Merck & Co.

Rahway, NJ

- Reduced error detection time from ~10 minutes to real time for a RAG application serving 50+ users by engineering an event-driven AWS pipeline that standardized frontend/backend errors and routed CloudWatch alerts through Lambda and SNS.
- Strengthened error-reporting reliability by updating React and FastAPI unit tests to validate service calls and payload integrity within the existing CI/CD pipeline.
- Drove cloud cost optimization across development teams by identifying unused environments and applications, helping teams achieve 95% AWS resource coverage.

Research Assistant

Sep. 2024 – Apr. 2026

Florida State University

Tallahassee, FL

- Co-authored a multi-agent LLM system for longitudinal behavioral health coaching, accepted for publication at IEEE/ACM CHASE 2026 (21% acceptance rate).
- Reduced redundant conversational looping by 55% (6.35% to 2.86%) by engineering a Python/vLLM multi-agent architecture with dedicated dialogue, state-tracking, and steering agents for longitudinal interactions.
- Designed a schema-validated state tracker and JSON parsing pipeline to maintain structured goals, barriers, and progress across multi-session conversations, enabling persistent cross-session memory.

Software Engineer Intern

Jun. 2025 – Aug. 2025

Merck & Co.

Rahway, NJ

- Streamlined quarterly audit planning for 20 auditors by developing a Power Apps and Power Automate solution that enabled teams to simulate audit schedules and coordinate assignments across multiple months.
- Automated audit scheduling and reminders by integrating Microsoft Graph with Outlook and Teams, synchronizing calendar events and notifications across auditors.

PROJECTS

Running Safer | runningsafer.com | *Next.js, React, TypeScript, Tailwind CSS, FastAPI, Python, PostgreSQL*

- Built and deployed a full-stack web app that imports Strava activities via OAuth and generates pedestrian-safety reports with interactive route maps.
- Engineered a route-analysis backend that maps Strava polylines to OpenStreetMap road and sidewalk data, scoring individual street segments on a 1–10 safety scale and highlighting areas of highest concern.

LEADERSHIP/INVOLVEMENT

Theta Tau Professional Engineering Fraternity | *Risk Chair*

Apr. 2024 – Present

- Led risk management for an 88-member chapter, organizing an FSUPD Chief of Police presentation on Title IX and sexual harassment alongside safety protocols for chapter events and a multi-day camping trip.

Association for Computing Machinery | *Historian*

Aug. 2024 – Apr. 2025

TECHNICAL SKILLS

Languages: C++, Python, C#, JavaScript, TypeScript, SQL, Bash, HTML, CSS

Frameworks/Libraries: React, Next.js, FastAPI, .NET, Blazor, Pytest, Vitest

Cloud/Databases: AWS Lambda, Amazon SNS, AWS CloudWatch, PostgreSQL

Developer Tools: Git, GitHub, Docker, Jira, Postman